



Evaluating natural resources volatility in an emerging economy: The influence of solar energy development barriers

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ABSTRACT

Utilizing natural resources aids in China's efforts to overcome its current energy issues. However, the country's ability to fulfill the rising energy demand is hindered by the instability of its natural resources. Multiple barriers to this failure have been identified by researchers. However, little is known about how these barriers affect the volatility of natural resources. In order to solve this research gap, the current work has used a systematic approach for ranking barriers according to their significance. Initially, we identified these barriers based on a comprehensive literature review and the modified Delphi method. Twenty-nine barriers were identified, which were subsequently grouped into six major dimensions, including technological, transparency & accountability, COVID-19, financial, policy & regulatory, and infrastructure. Secondly, using the Analytical Hierarchical Process (AHP), the relative weights and ranks of major barriers and sub-barriers were calculated. The results demonstrate that the "technological" barrier is the highest-ranked barrier, while "unreliable local technology" rates highest among all sub-barriers. Finally, we took an additional step by utilizing the Grey Preference by Similarity to Ideal Solution (G-TOPSIS) method to rank the alternatives. The results indicate that "R&D" and "corruption-free system" are the most viable alternatives for overcoming these barriers. Research findings emphasize ensuring the availability of a skilled workforce, reforming policy structure, developing infrastructure, improving R&D activities, and highlighting the paybacks that solar energy offers to preserve natural resources by integrative and consistent efforts.

1. Introduction

Elevated carbon emissions due to the burning of fossil fuels are damaging the environment (Anser et al., 2020b). In September 2015, the United Nations unveiled 17 Sustainable Development Goals (SDGs), which were endorsed by all United Nations member states to attain global peace and prosperity (Awan et al., 2021a; Nassani et al., 2019). As a developing country, fossil fuels are China's principal source of energy generation (Sun et al., 2022). These sources satisfy most of the country's energy demands (Cheng et al., 2021). The shortage of fossil fuels is exacerbated by their extensive consumption (Razzaq et al., 2022a). Due to its heavy reliance on these sources, future fossil fuel depletion is expected for China. In the meantime, the country's power

consumption and supply situation have worsened due to the massive population and economic expansion. China is seeking alternative energy sources to minimize its reliance on fossil fuels to cope with the current climate crisis (Fang et al., 2022). The Chinese government has proposed several programs to increase the proportion of renewable energy sources in the country's overall energy mix. In addition to enhancing the country's energy structure, solar energy can reduce carbon footprints and promote environmental sustainability (Azam et al., 2021). As a consequence of government measures, entrepreneurs and investors have shown a great deal of interest in solar energy development projects, and new solar parks have been constructed in various areas across the country (Pouran, 2018).

Despite the country's abundant solar resources, China's solar energy

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deployment is impeded by a number of barriers (Wu et al., 2018). Scholars concurred that the fate of China's solar energy potential would stay unchanged until these barriers are thoroughly recognized and debated. In order to develop individualized solutions, it is necessary to identify, assess, and rank these barriers according to their relevance. Multiple studies have highlighted geographical and temporal constraints associated with renewable energy sources. Some of them even offered optimal solutions in a variety of circumstances (Luthra et al., 2015; Shah et al., 2019). However, depending on the economic, social, political, and technical climates, these barriers may differ in almost every country. For example, Irfan et al. (2019) evaluated Pakistan's solar energy barriers. Major impediments were classified by the authors using semi-structured interviews with industry specialists. The research identified twenty-three barriers in five broad sectors, including technological, economical, policy, social, and information & human resources. The authors placed economical and policy barriers at the top of the list, followed by technological, information & human resources, and social categories.

Employing the Analytic Hierarchy Process (AHP) methodology, Numata et al. (2020) examined barriers to the implementation of renewable-based micro-grids in Myanmar. The authors identified twenty barriers and categorized them into five primary areas. Economic, technical, and regulatory barriers substantially affect the implementation of micro-grids, whereas social and financial barriers have a lesser effect. Mukeshimana et al. (2021) analyzed biogas distribution barriers in the Rwandan context using the AHP methodology. In total, twenty of the most important barriers were found within four broad dimensions. According to the authors, financial barriers are the most important, followed by technical, sociocultural, and institutional impediments.

Few researchers have identified and categorized solar energy development barriers in the Chinese context. However, despite the enduring interest of previous researchers, there are few studies on the extent to which these barriers inhibit the volatility of China's natural resources, especially during the COVID-19 epidemic. Prior to formulating national policy, it is of the highest necessity to comprehend the scope of impediments. The lack of such studies prompted us to fill this research gap by performing a comprehensive study and answering the following three research questions. (i) What are the primary barriers to solar energy development in China? How might they be prioritized in a way that benefits both scholars and practitioners? (iii) What are the most effective solutions for overcoming these barriers to the sustainability of natural resources in the country? Government and policy-makers will be in a better position to combat natural resource volatility if they become aware of the impact of solar energy development barriers. For this purpose, we utilize the AHP approach to calculate the weights of the barriers. As a next stage, the G-TOPSIS method is utilized to determine the ultimate ranking of the closest alternative to the best solution. Flexibility, rationality, and understandability are the primary advantages of the G-TOPSIS methodology, enabling the comparative accomplishment of options to be computed in a computational manner. This study made threefold contributions. In the first place, unlike other scholars, we have utilized the modified Delphi method in the Chinese solar energy setting. Secondly, the study offers contributions in the form of novel insights concerning solar energy development barriers by including, for the very first time, the pandemic (COVID-19) as a new barrier. To the best of the authors' knowledge, this barrier has not previously been utilized in the context of solar energy. In addition, when we revealed fresh insights into significant sub-criteria, we identified AHP results as a foundation for identifying main barriers and sub-barriers for researchers and policymakers. Lastly, further contributions are made by implementing G-TOPSIS, which would aid in expanding past findings, selecting the best solutions to overcome natural resources volatility in the country, and confirming the coherence of study outcomes.

The remaining research procedures are as follows: Section 2 defines the paper's research framework. Section 3 provides results and

discussion. The paper concludes with section 4, along with proposing key policy recommendations.

2. Research framework

Fig. 1 depicts the research design of this study. As a preliminary step, a thorough literature analysis is undertaken to identify the major barriers to solar energy. Journal papers, energy studies, authorized government policy documents, and regulatory reports were extensively reviewed to better understand the barriers. The Web of Science, Scopus, Google Scholar, ScienceDirect, and IEEE Explore databases were searched for scholarly journal papers. Various agencies, including the International Renewable Energy Agency (IRENA) and the World Bank, were surveyed for their published energy reports. The majority of policy and regulatory reports were collected from different organizations, including the National Bureau of Statistics, China Statistical Yearbook, China Science and Technology Statistical Yearbook, Urban Statistical Yearbook, the website of the Ministry of Industry and Information Technology of China, and China High-tech Industry Statistical Yearbook. Next, the modified Delphi technique is utilized to finalize and categorize solar energy barriers with the assistance of experts' feedback. Later, the AHP method is used in to determine the relative weights of barriers by constructing a pairwise comparison matrix. Finally, the G-TOPSIS method is then employed to determine the optimal solution to these barriers. In the respective sections, a thorough description of these methods is provided.

2.1. Modified Delphi method

For the classification of solar energy barriers, the modified Delphi approach is utilized. Using discussions, this approach systematically collects feedback from experts. By communicating their ideas, experience, views, and abilities, experts from related domains are able to make common inferences (Revez et al., 2020). The process consists of five important elements, including the selection of experts, a double round of questionnaire surveys in the second and third steps, the gathering of experts' feedback, and the rating of barriers (Quyên, 2014).

According to the literature, there is no unique criterion about the number of experts utilized in this strategy. However, Shah et al. (2020) recruited sixteen experts in their post-COVID-19 green economic recovery research. Zare et al. (2018) solicited opinions from five experts, whereas Solangi et al. (2020) polled twelve experts for their analyses. Due to the nationwide lockdown situation, we performed an online discussion with 15 professionals (from academia and industry) and integrated their feedback using the modified Delphi methodology. Specifically, we gathered data from specialists during 15 April – 30 May 2020. There were three steps to the data gathering procedure. During the first step, 17 specialists were contacted through email and requested for permission to participate in the study. At this point, they were given a brief explanation of the study's objectives. The following criteria were considered for the recruitment of specialists: (1) Top-tier experts from academia and industry. (2) They should have a minimum of 10 years of expertise in their respective professions. (3) They must have a bachelor's degree or above in order to be considered qualified for the survey. In consideration of the significant roles that barriers can play in impeding solar energy development in China, fifteen experts consented to share their opinions. Based on previous research, the sample size (15 experts) is enough to undertake this study. The second step provided a one-week time for agreed experts to deliver their feedback. A comprehensive explanation of every question was supplied for accurate and relevant findings. During the third step, feedback was collected from experts and incorporated into the empirical analysis. Experts supplied scores based on grey linguistic factors and triangular fuzzy numbers (TFN) in order to overcome solar energy barriers. Appendix A contains the profiles of the experts.

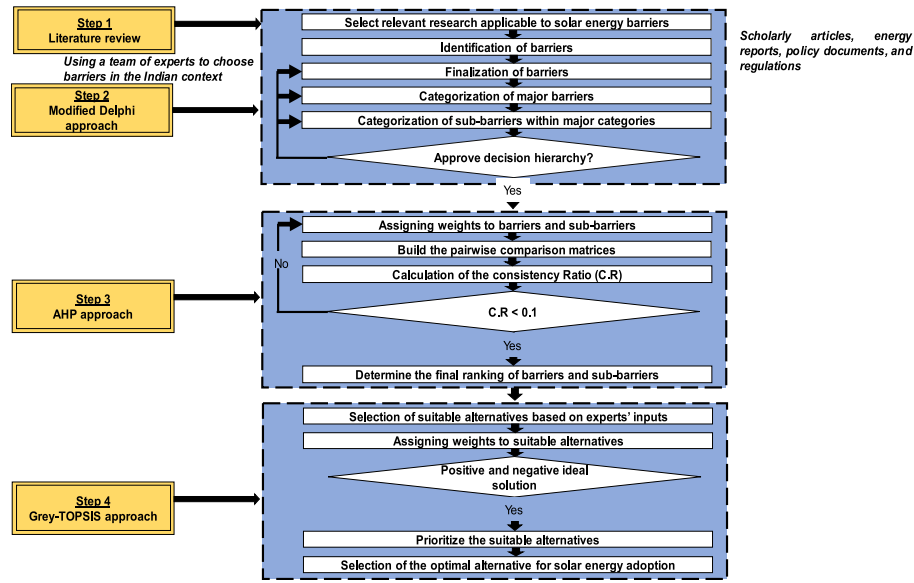


Fig. 1. Research framework.

2.2. Analytical hierarchy process

AHP is amongst the most versatile MCDM strategies for solving difficult decision-making issues. In the 1970s, Thomas L. Saaty was the first to propose this method (Saaty, 1977). As per Wang et al. (2020), AHP's major components include pairwise comparisons, therefore generating comparison matrices whose correctness may be verified and maintained appropriately. The strategy aids decision-makers in accurately selecting and ranking difficult issues. Saaty (1990) outlined the primary steps of the AHP technique as follows:

Step 1 Establishment of the problem's hierarchical structure.

Step 2 Development of a pairwise comparison matrix for a decision issue by utilizing Saaty's 1-9-point scale (see Table 1).

Step 3 The Consistency Index (CI) computation. This step computes the consistency of the pairwise comparison matrix using CI as:

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (1)$$

Here, λ_{max} specifies the eigenvalue, and n specifies the no. of major criteria.

Phase 4 The Consistency Ratio (CR) computation.

$$CR = \frac{CI}{RI} \quad (2)$$

Where, RI represents the random index, which is presented in Table 2. A primary requirement that must be addressed during pairwise comparisons is that CR must fall inside the recommended value of 0.1. If it exceeds the recommended level, the findings will be erroneous (Contreras-Nieto et al., 2019). The AHP methodology assigns weights to

the major barriers and sub-barriers when all phases have been completed.

2.3. G-TOPSIS approach

The TOPSIS methodology was introduced by Yoon and Hwang (1985). Using the m arguments, the algorithm evaluates n options. The fundamental logic of this strategy establishes the ideal positive and negative solutions. An optimal solution has a minimum distance to the ideal positive solution and a maximum distance to the ideal negative solution. The following steps outline the distinct phases of the G-TOPSIS methodology (Ikram et al., 2020).

Step 1 Decision-makers determine the relative importance of each criterion in this step. Using linguistic factors, the grey theory may express the perspectives of experts. The weight of criteria j is calculated by the following Eq. if k signifies decision-makers.

$$\otimes w_j = \frac{1}{k} \left[\otimes w_j^1 + \otimes w_j^2 + \dots + \otimes w_j^k \right] \quad (3)$$

Step 2 The linguistic factors are applied to determine the relative position of the research options in relation to the criterion. If k indicates decision-makers for criteria j , then the value of alternative i may be calculated as follows:

$$\otimes G_{ij} = \frac{1}{k} \left[\otimes G_{ij}^1 + \otimes G_{ij}^2 + \dots + \otimes G_{ij}^k \right] \quad (4)$$

Step 3 This step is described by the following Eq. and involves the grey decision matrix.

$$D = \begin{bmatrix} \otimes G_{11} & \otimes G_{12} & \dots & \otimes G_{1n} \\ \otimes G_{21} & \otimes G_{22} & \dots & \otimes G_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \otimes G_{m1} & \otimes G_{m2} & \dots & \otimes G_{mn} \end{bmatrix} \quad (5)$$

Here, $\otimes G_{ij}$ articulates the significance of the alternative i in the criterion j .

Table 1

Saaty's AHP scale.

Number	Linguistic scale
1	Equal importance (HU)
3	Moderate importance (U)
5	Strong importance(N)
7	Very strong importance (F)
9	Extreme importance (HF)
2,4,6,8	Intermediate values between adjacent scale values

Table 2
Random index (RI).

n	1	2	3	4	5	6	7	8	9	10
RI	0.00	0.00	0.058	0.90	1.12	1.24	1.32	1.41	1.45	1.49

Step 4 In this stage, the matrix of grey decision is standardized and denoted by Equation (6) as:

$$D^* = \begin{bmatrix} \otimes G_{11}^* & \otimes G_{12}^* & \dots & \otimes G_{1n}^* \\ \otimes G_{21}^* & \otimes G_{22}^* & \dots & \otimes G_{2n}^* \\ \vdots & \vdots & \vdots & \vdots \\ \otimes G_{m1}^* & \otimes G_{m2}^* & \dots & \otimes G_{mn}^* \end{bmatrix} \quad (6)$$

If the criterion is benefit characteristics, then Normalization Eq. (7) is used as:

$$G_{ij}^* = \left[\frac{G_{ij}}{G_j^{\max}}, \frac{\bar{G}_{ij}}{G_j^{\max}} \right] \text{ Where } G_j^{\max} = \max_{1 \leq j \leq m} \{ \bar{G}_{ij} \} \quad (7)$$

For normalization, the following Eq. is utilized when the criterion is cost characteristics.

$$G_{ij}^* = \left[\frac{G_j^{\min}}{G_{ij}}, \frac{G_j^{\min}}{\bar{G}_{ij}} \right] \text{ Where } G_j^{\min} = \min_{1 \leq j \leq m} \{ \underline{G}_{ij} \} \quad (8)$$

Following normalization, the values of the energy matrix would fall inside the interval [0, 1].

Step 5 The following Eq. depicts the development of a grey-weighted normalized decision matrix.

$$V = \begin{bmatrix} \otimes V_{11} & \otimes V_{12} & \dots & \otimes V_{1n} \\ \otimes V_{21} & \otimes V_{22} & \dots & \otimes V_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ \otimes V_{m1} & \otimes V_{m2} & \dots & \otimes V_{mn} \end{bmatrix} \text{ Where } \otimes V_{ij} = \otimes G_{ij}^* \times \otimes w_j \quad (9)$$

Step 6 The optimum positive and negative solutions are calculated using the following Equations:

$$S^{\max} = \left\{ \left[\max_{1 \leq j \leq m} \underline{V}_{i1}, \max_{1 \leq j \leq m} \bar{V}_{i1} \right], \left[\max_{1 \leq j \leq m} \underline{V}_{i2}, \max_{1 \leq j \leq m} \bar{V}_{i2} \right], \dots, \left[\max_{1 \leq j \leq m} \underline{V}_{in}, \max_{1 \leq j \leq m} \bar{V}_{in} \right] \right\} \quad (10)$$

$$S^{\min} = \left\{ \left[\min_{1 \leq j \leq m} \underline{V}_{i1}, \min_{1 \leq j \leq m} \bar{V}_{i1} \right], \left[\min_{1 \leq j \leq m} \underline{V}_{i2}, \min_{1 \leq j \leq m} \bar{V}_{i2} \right], \dots, \left[\min_{1 \leq j \leq m} \underline{V}_{in}, \min_{1 \leq j \leq m} \bar{V}_{in} \right] \right\} \quad (11)$$

Step 7 During this step, Eq. (12) is used to determine the probable greyness degree between ideal and alternative solutions.

$$P\{S_i \leq S^{\max}\} = \frac{1}{n} \sum_{j=1}^n P\{ \otimes V_{ij} \leq \otimes G_j^{\max} \} \quad (12)$$

Step 8 Alternatives are ordered ascendingly based on the values acquired in the previous step. The alternative with the lowest degree of greyness is given higher attention.

3. Results and discussion

3.1. Results of the AHP approach

The analysis begins by conducting a comprehensive literature review. Next, a group discussion technique of decision-making was employed to give weights to solar energy barriers and sub-barriers. Based on an extensive literature review and experts' inputs, six major barriers were identified for the AHP model analysis. These barriers include technological, transparency & accountability, COVID-19, financial, policy & regulatory, and infrastructure. Next, we performed a pairwise AHP comparison matrix to measure individual weight scores by utilizing a geometric mean (X. Zhou et al., 2018). In the first step of AHP evaluation, the weights of the major criteria were determined, while twenty-nine sub-barriers were evaluated in the second phase. Fig. 2 depicts the hierarchical structure of solar energy barriers.

3.1.1. Ranking of major barriers

The AHP technique determined the relative importance of all major barriers using a hierarchical weighting system (see Table 3). It is evident from Table 3 that the technological barrier is the most influential barrier, having a weight of (0.383). Transparency & accountability barrier ranks second with a weight of (0.260), followed by COVID-19 (0.143), financial (0.097), policy & regulatory (0.061), and infrastructure (0.056) barriers respectively. Fig. 3 represents the detailed weights of each major barrier. Based on the weight estimations of the major barriers, the top priority is given to the technological barrier in this study. Developing intelligent manufacturing systems for eco-friendly production is a top goal for businesses (Anser et al., 2020a). However, solar technology is still associated with high risks in China. The second priority is given to the transparency & accountability barrier. Lack of implementation of transparent rules and regulations, unchecked bureaucracy powers, high taxation, and the monopoly of government institutions are the root causes of the transparency & accountability

barrier (Tyfield et al., 2015).

The recent COVID-19 pandemic is the third most significant barrier to the solar energy sector, impacting the overall solar industry and the solar energy development projects (Eroglu, 2020). The financial barrier turned out to be the fourth most influencing barrier after COVID-19. Owing to the latest advancements, new guidelines and enforcement parameters continue to be significant problems for the solar industry. The industry, therefore, needs to regularly adopt new protocols to allow more sustainable practices and comply with them. The solar sector is also confronting policy & regulatory barriers. The absence of an integrated solar energy policy and regulatory mechanisms affects the rapid

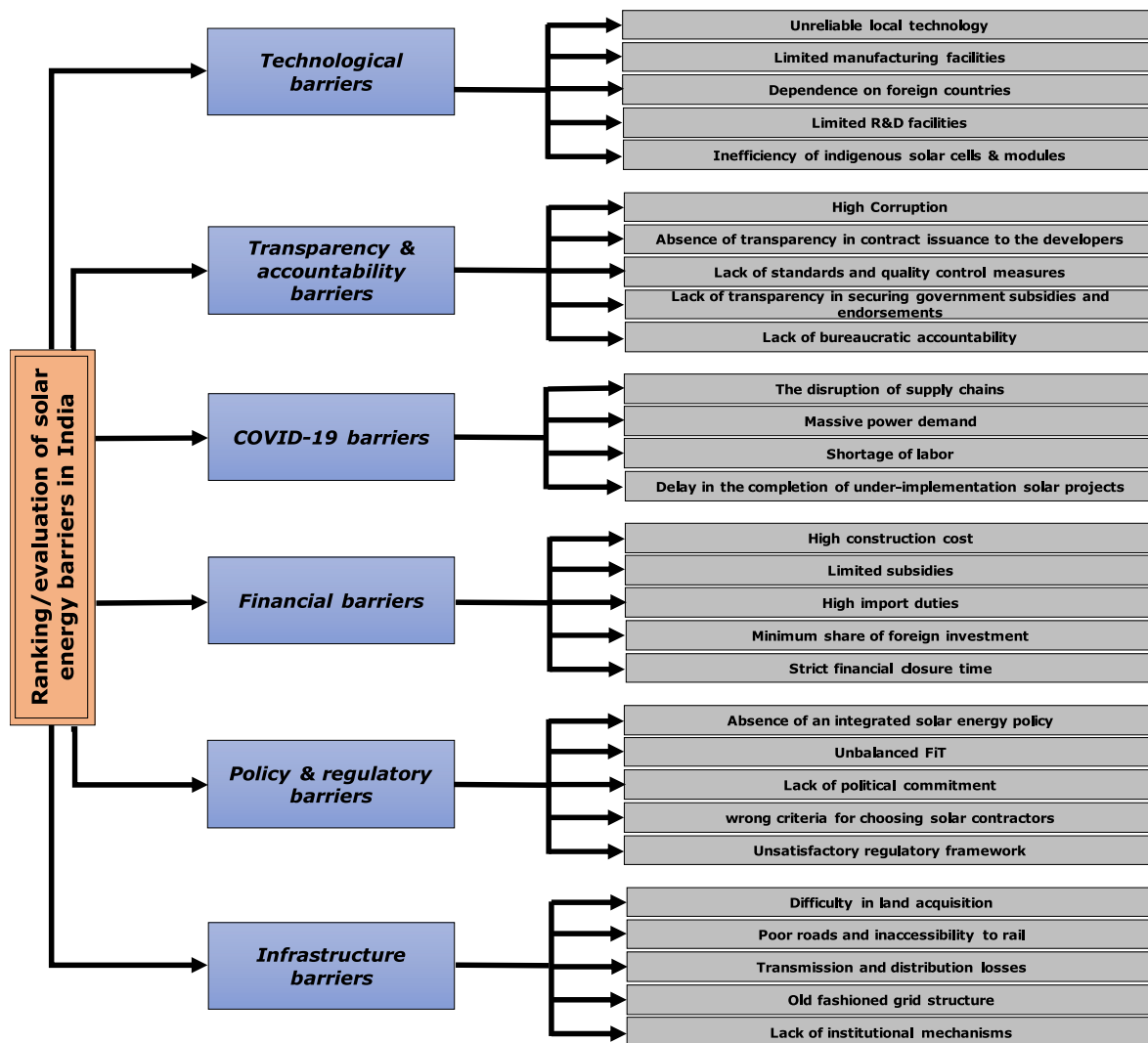


Fig. 2. The hierarchal structure of solar energy barriers.

Table 3
Ranking of major barriers.

Major barriers	Code	Weight	Rank
Technological	TECH	0.383	1st
Transparency & accountability	TRA	0.260	2nd
COVID-19	COV	0.142	3rd
Financial	FIN	0.097	4th
Policy & regulatory	POLR	0.061	5th
Infrastructure	INF	0.056	6th

CR = 0.064.

development of the solar industry in China. Besides, some Government policies (i.e., ¹RPO, FiT, accelerated depreciation, and net-metering policy) need amendments with the passage of time. Finally, the least significant barrier is the infrastructure barrier, which obtained the least weight estimated by the fuzzy AHP method in the context of solar energy, impeding the continuous deployment of solar energy in the country.

3.1.2. Ranking of sub-barriers

After building the hierarchical structure in Appendix B, a pairwise

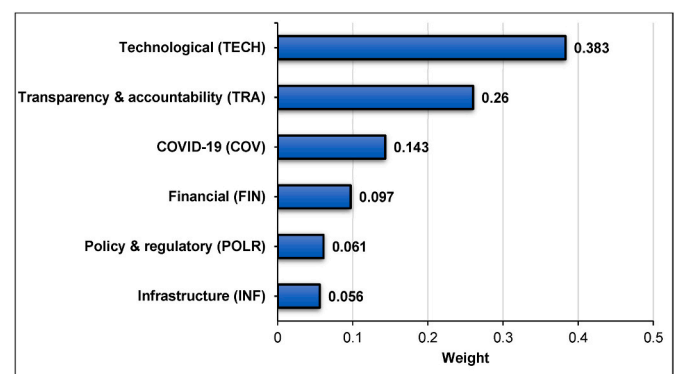


Fig. 3. The ranking of major barriers.

comparison matrix of sub-barriers is constructed (see Tables B1–B6). The results of sub-barriers are discussed in this section.

3.1.2.1. Technological. According to the results shown in Fig. 4, the sub-barriers in the technological category are ranked as follows: TECH-4 > TECH-5 > TECH-2 > TECH-1 > TECH-3. Unreliable local technology (TECH-4) is by far the most critical barrier in this domain, with a

¹ RPO: Renewable purchase obligations, FiT: Feed-in tariffs.

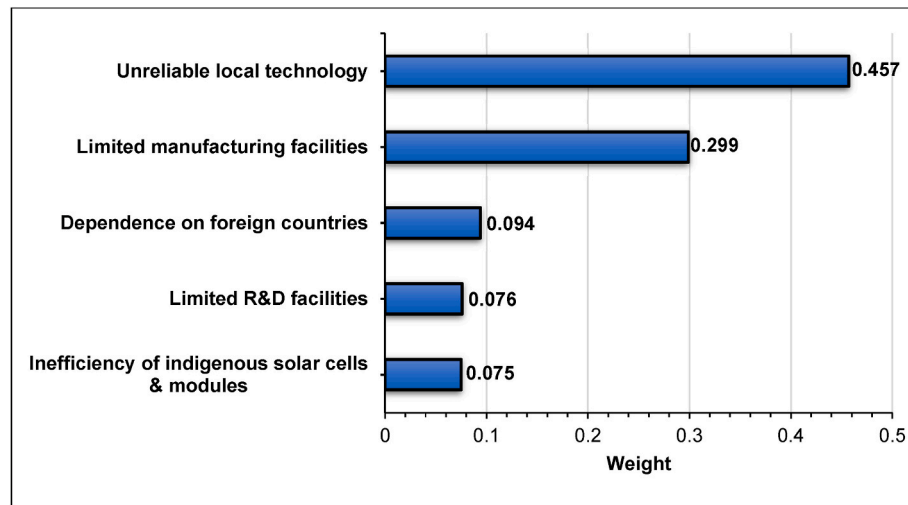


Fig. 4. The ranking of technological sub-barriers.

maximum weight of 0.457. As a result of the industrial revolution's automation, digital technologies have connected everything and everyone in order to collect data for analysis (Tang et al., 2022), display the whole production system, and offer relevant data to the business system (Awan et al., 2021b). However, solar technology is not so developed in China, and hence there are high risks associated with this technology. The availability of solar irradiation data is necessary for the establishment of solar parks. However, some developers in the country merely use data obtained from the National Renewable Energy Laboratory (NREL), satellite modeled data (with only 70–80% confidence level) of the National Aeronautics and Space Administration (NASA), or assign external organizations to determine on-ground data. The dispatch of newly established solar parks is below projections due to the unavailability of proper solar irradiation data (Li et al., 2011).

Limited manufacturing facilities (TECH-5) ranked second by obtaining a weight of 0.299. Literature supported this fact as almost every advanced country is developing its solar sector by building large manufacturing facilities. Countries such as the USA, Malaysia, and Germany have already achieved multi-gigawatt productions; however, the manufacturing capacity of solar cells and modules is relatively low in China. Dependence on foreign countries (TECH-2) obtained a weight of 0.094 and is the third critical barrier in this category. Due to unreliable local technology and limited manufacturing facilities, China relies on foreign countries for key equipment used in the construction of solar farms (Zhang et al., 2010). The import of these components from foreign countries not only results in high costs but also slows down the domestic

growth of the solar industry.

Limited research and development (R&D) facilities (TECH-1) ranked fourth in the category with 0.076 weight. The solar industry is not much developed in China compared to other renewable industries. To remain competitive and for the speedy growth of the solar sector, the country needs to invest more in R&D facilities on a priority basis (Zhao et al., 2013). The inefficiency of indigenous solar cells and modules (TECH-3) obtained a weight of 0.075 and got last place under the technological sub-barrier category.

3.1.2.2. Transparency & accountability. Transparency & accountability sub-barriers are ranked as follows: TRA-4 > TRA-3 > TRA-2 > TRA-5 > TRA-1 and presented in Fig. 5. The top sub-barrier in this dimension is high corruption (TRA-4), as it obtained the highest weight of 0.418. Researchers have widely reported this problem in their studies. For instance, Ali and Gasmi (2017) exposed that corruption is a primary barrier to economic development. Spending on government projects and schemes is the main corruption source in the country. Developers are forced to give bribes to officials during all phases of solar energy projects. The absence of transparency in contract issuance to the developers (TRA-3) is the second critical sub-barrier within this dimension, with 0.249 wt

Lack of standards and quality control measures (TRA-2) obtained a weight of 0.141 and got third place in this category. Lack of transparency in securing government subsidies and endorsements (TRA-5) ranks fourth in this dimension, with 0.104 weight. Local developers in

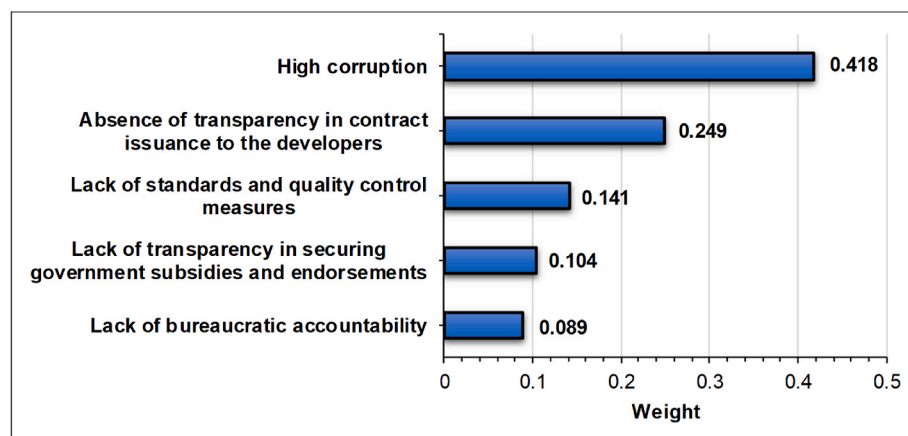


Fig. 5. The ranking of transparency & accountability sub-barriers.

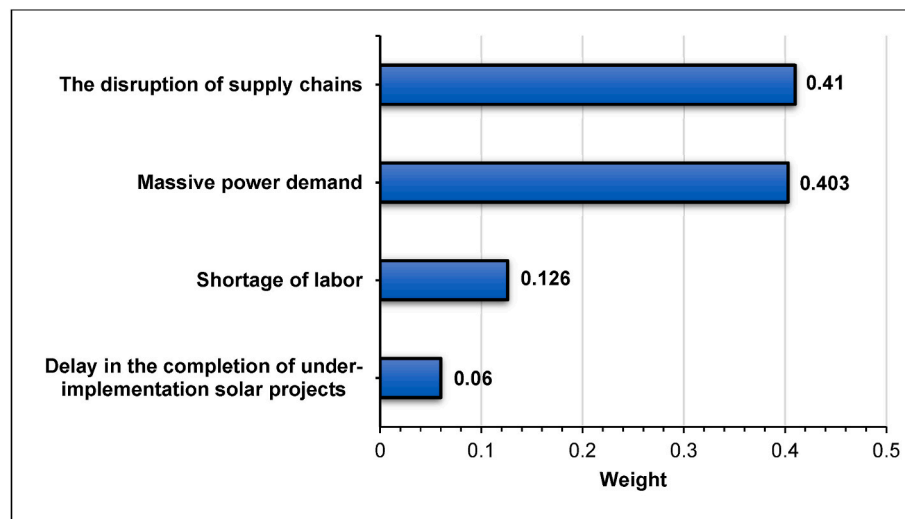


Fig. 6. The ranking of COVID-19 sub-barriers.

China think that investing in solar energy development projects is difficult in the country compared to foreign countries. The major reason is the government's lack of transparency in getting endorsements and financial assistance. Consequently, local developers prefer to invest in foreign countries (García, 2013). Last in this category is the lack of bureaucratic accountability (TRA-1), with a weight of 0.089.

3.1.2.3. COVID-19. The results of the COVID-19 sub-barrier are shown in Fig. 6. The ranking of sub-barriers under COVID-19 (COV) group is as follows: COV-2 > COV-1 > COV-3 > COV-4. The disruption of supply chains (COV-2) is identified as the most significant COVID-19 sub-barrier with a weight of 0.410. The ranking is reasonable as governments have imposed strict restrictions in the form of quarantine and social distancing policies (Razzaq et al., 2022b). Like other businesses, the supply chains of the solar industry have been severely influenced due to these restrictions. Accordingly, the production facilities of solar parks and power plants were considerably reduced (Pradhan et al., 2020). To remain competitive, it is essential to gather knowledge resources (Awan et al., 2020). Massive power demand (COV-1) placed second within this domain by securing 0.403 weight. China is the largest country in terms of population. The novel pandemic has changed the lifestyles of people, as now people typically stay at home and work. Hence, there is a significant increase in residential power demand. This terrible situation poses new challenges for the power sector in the country (Yang et al., 2021).

Shortage of labor (COV-3) obtained a weight of 0.126 and ranked third in the COVID-19 sub-barrier category. The solar industry is facing

immediate challenges as a consequence of COVID-19, such as the shortage of labor. The current labor force situation is in disarray due to the lockdown in different Chinese provinces, which has resulted in a labor crunch, and the solar industry has not been left untouched by this unprecedented crisis (Choi and Brindley, 2021). One of the key contributors to the government's attempt to attain clean energy goals, the solar sector, has been left high and dry. Delay in the completion of under-implementation solar projects (COV-4) is placed in the fourth position, as it received a weight of 0.060. The completion of under-implementation solar projects in China has been delayed, as China has closed borders with other countries and canceled all consignments for the time being, causing significant delays in the completion of ongoing solar projects (Wang et al., 2021).

3.1.2.4. Financial. Financial (FIN) sub-barriers are ranked as follows: FIN-4 > FIN-5 > FIN-1 > FIN-2 > FIN-3 (see Fig. 7). High construction cost (FIN-4) is the top sub-barrier within this dimension with 0.408 weight. Previous literature also supported this argument, as Liu (2018) revealed in their study that high cost is a major barrier in solar energy development because the construction of solar energy parks requires high capital cost. Though the cost of solar energy technology has considerably reduced over the years, it is still high in developing countries like China, which supports the use of fossil fuel-based thermal energy (Irfan et al., 2021).

Based on weight allocation, limited subsidies (FIN-5) is the second most significant sub-barrier having a weight of 0.386. Contrary to thermal electricity, solar energy has no confirmed track record. Consequently, it is difficult for new ventures to receive subsidies from the government. Monetary organizations have limited investment exposure against solar energy due to the fact that a vast capital has already been extended for thermal power sources (Bai et al., 2018). As a result, the solar sector is confronted with teething pains. High import duties (FIN-1) obtained 0.091 wt and took third place in this category. Solar energy programs have become uneconomical due to the high import duties. The government has imposed heavy duties on the import of solar PV, which has resulted in the high capital costs of PV plants (WU, 2014). The minimum share of foreign investment (FIN-2) with a weight of 0.064 is the fourth significant barrier under this category. As the interest rate for solar power programs is very high, foreign developers are reluctant to invest in the Chinese solar sector (Cherni and Kentish, 2007). Strict financial closure time (FIN-3) turned out to be the least important sub-barrier within this domain, with a weight of 0.051.

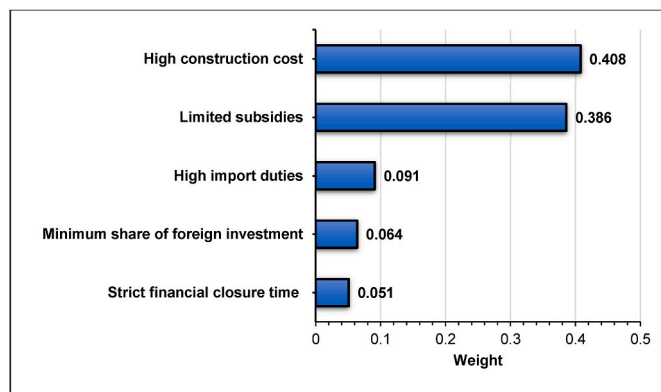


Fig. 7. The ranking of financial sub-barriers.

3.1.2.5. Policy & regulatory. The ranking of sub-barriers under 'policy &

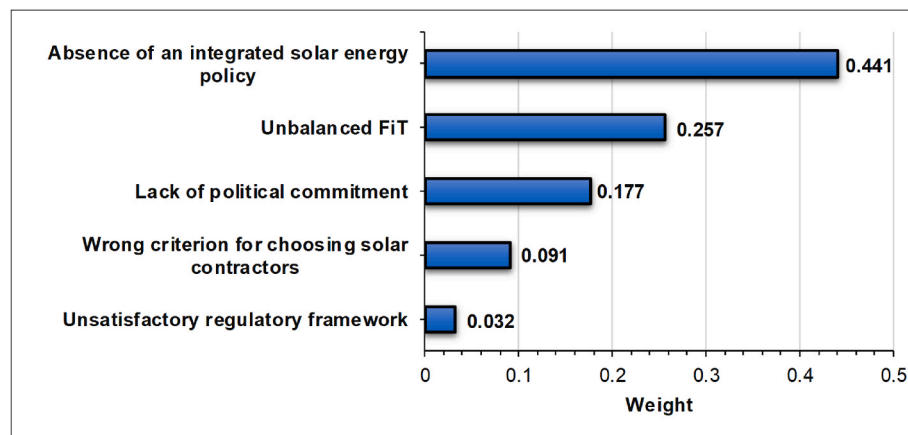


Fig. 8. The ranking of policy & regulatory sub-barriers.

regulatory' (POLR) category is as follows: POLR-2 > POLR-5 > POLR-4 > POLR-3 > POLR-1 as depicted in Fig. 8. The absence of an integrated solar energy policy (POLR-2) is the most significant sub-barrier with a weight of 0.441 in this list. Various studies have reported that the absence of explanation on existing and forthcoming guidelines for the growth of solar power capacity, infrastructure, and the establishment of grids continues to deter stakeholders (Behuria, 2020; Hairat and Ghosh, 2017). Unbalanced FiT (POLR-5) placed second in the category with 0.257 weight. The government has specified FiT in all provinces. Nevertheless, FiT, governed by a cost-plus strategy within a rate of return mechanism, restrains the full use of solar power and never provides the cost reduction benefits. This prevalent form of cost-plus FiT has severely overlooked economic efficiency (Zhao et al., 2016).

Lack of political commitment (POLR-4) obtained a weight of 0.177 and secured third place in this domain. China has abundant solar energy potential. Without a robust political commitment, this potential cannot be realized. Government intrusion in local markets are hindering the deployment of solar energy in the country (Keeley and Matsumoto, 2018). Wrong criteria for choosing solar contractors (POLR-3) is next in the sub-barriers' category with 0.091 weight. Currently, the benchmark for selecting solar contractors is based on minimum to maximum capability instead of track record and financial status (Yan et al., 2019). With a weight of 0.032, Unsatisfactory regulatory framework (POLR-1) got last place in the category of policy & regulatory barriers.

3.1.2.6. Infrastructure. The ranking of sub-barriers within the 'infrastructure' (INF) group, exhibit the following pattern as: INF-2 > INF-5 > INF-4 > INF-3 > INF-1 (see Fig. 9). Difficulty in land acquisition (INF-2) is the top sub-barrier under the infrastructure domain, obtaining a weight of 0.435. Land acquisition for setting solar energy projects is always a lengthy and time-consuming process. Nearly one year is needed

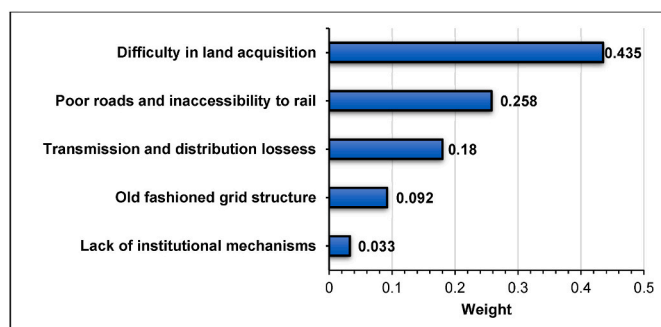


Fig. 9. The ranking of infrastructure sub-barriers.

to acquire land, and in some circumstances, even one year is insufficient. Approval from different Government departments is also required for this purpose. Due to these difficulties, many solar energy projects have become delayed or even relocated to other places (Keeley and Matsumoto, 2018).

Poor roads and inaccessibility to rail (INF-5) rank second in the category by obtaining 0.258 weight. Sometimes, solar parks are established in remote rural areas, far away from national highways, and do not have accessibility to rail. In these conditions, the transportation of PVs and related equipment to the target site is a challenging task. Even in some cases, the costly solar power equipment breaks during transportation, which results in delaying the normal function of solar power development programs (Sindhu et al., 2017). Transmission and distribution (T&D) losses (INF-4) is the third major sub-barrier in this category, as it has obtained a weight of 0.180. Poor management, less monitoring, and electricity theft are the common reasons for T&D losses. The national grid is China's high-voltage electricity transmission network, connecting power plants and main substations and assuring that electricity generated anywhere in the country can be used to fulfill demand in other parts of the country (L. Zhou et al., 2018).

Old fashioned grid structure (INF-3) secured 0.092 wt and placed fourth under the infrastructure barrier category. Literature supported this fact, as the China grid structure is old-fashioned and is built for conventional thermal energy. Consequently, it only promotes thermal energy. Solar parks lack grid infrastructure in the country. Sun (2020) reported that the existing grid structure needs massive investment to support renewable energy technologies, including solar energy, in the long run. Finally, the lack of institutional mechanisms (INF-1) got the last position in the list by obtaining a weight of 0.033. There are no institutional mechanisms to offer after-sales services to support solar technology. Limited involvement of the private sector and associated initiatives cannot inspire the present institutional tools to adapt to new markets (Sheng, 2020).

3.1.3. Overall ranking of sub-barriers

We determined the overall weights of sub-barriers by multiplying each sub-category barrier's weight by its priority weight (see Table 4). Fig. 10 shows the schematic representation of the overall ratings of solar energy sub-barriers. The criterion is as follows: TECH-4 > TECH-5 > TRA-4 > TRA-3 > COV-2 > COV-1 > FIN-4 > FIN-5 > TRA-2 > TECH-2 > TECH-1 > TECH-3 > TRA-5 > POLR-2 > INF-2 > TRA-1 > COV-3 > POLR-5 > INF-5 > POLR-4 > INF-4 > FIN-1 > COV-4 > FIN-2 > POLR-3 > INF-3 > FIN-3 > POLR-1 > INF-1.

It is evident from Fig. 10 that "unreliable local technology" ranks highest among all sub-barriers by obtaining a weight of 0.1750, followed by "limited manufacturing facilities" (0.1145). "High corruption" and

Table 4
Final weights and ranking of sub-barriers.

Major barriers	Major criteria weights	Sub-criteria' codes	Priority weight	Final weight	Final ranking
Technological (TECH)	0.383	TECH -1	0.076	0.0291	11
		TECH -2	0.094	0.0360	10
		TECH -3	0.075	0.0287	12
		TECH -4	0.457	0.1750	1
		TECH -5	0.299	0.1145	2
Transparency and accountability (TRA)	0.260	TRA -1	0.089	0.0231	16
		TRA -2	0.141	0.0367	9
		TRA -3	0.249	0.0647	4
		TRA -4	0.418	0.1087	3
		TRA -5	0.104	0.0270	13
COVID-19 (COV)	0.143	COV-1	0.403	0.0576	6
		COV-2	0.410	0.0586	5
		COV-3	0.126	0.0180	17
		COV-4	0.060	0.0086	23
Financial (FIN)	0.097	FIN-1	0.091	0.0088	22
		FIN-2	0.064	0.0062	24
		FIN-3	0.051	0.0049	27
		FIN-4	0.408	0.0396	7
		FIN-5	0.386	0.0374	8
Policy and regulatory (POLR)	0.061	POLR -1	0.032	0.0020	28
		POLR -2	0.441	0.0269	14
		POLR -3	0.091	0.0056	25
		POLR -4	0.177	0.0108	20
		POLR -5	0.257	0.0157	18
Infrastructure (INF)	0.056	INF -1	0.033	0.0019	29
		INF -2	0.435	0.0244	15
		INF -3	0.092	0.0052	26
		INF -4	0.180	0.0101	21
		INF -5	0.258	0.0145	19

“the absence of transparency in contract issuance to the developers” are the next important sub-barriers by securing 0.1087 and 0.0647 wt, respectively. The old-fashioned grid structure (0.0052), minimum share of foreign investment (0.0049), unsatisfactory regulatory framework (0.0020), and the lack of institutional mechanisms (0.0019) are the least critical sub-barriers.

3.2. Results of the G-TOPSIS approach

The G-TOPSIS method improves the accuracy of decision-making in ambiguous circumstances. Consequently, we have employed this approach and selected the optimal solution to eradicate the barriers to the successful adoption of solar energy in the country. A group of 15 experts provided their inputs regarding the key alternatives to overcome the solar energy adoption barriers. The five key alternatives, such as human resources availability (A1), R&D improvement (A2), policy reforms (A3), corruption-free system (A4), and infrastructure development (A5), were chosen. Based on experts' inputs, we used the decision-making approach to determine the results. The inputs from the group of experts determined grey numbers and a grey evaluation matrix utilizing linguistic variables. After comparing alternatives to each sub-barrier category, the ranking matrix of linguistic characteristics was then created. Table 5 displays the findings of the positive grey ideal solution and the negative fuzzy ideal solution derived using G-TOPSIS. Finally, we assess the degree of greyiness between ideal solutions and alternatives. Eq. (13)–(18) illustrate the optimal alternatives

$$P(S_1 \leq S^{max}) = 0.401 \quad (13)$$

$$P(S_2 \leq S^{max}) = 0.230 \quad (14)$$

$$P(S_3 \leq S^{max}) = 0.512 \quad (15)$$

Table 5
Priority of alternatives based on G-TOPSIS.

Codes	Alternatives	S^{max}	S^{min}	possibility degree of greyness	Rating
A1	Human resources availability	0.495201	0.403214	0.401	3rd
A2	R&D improvement	0.61266	0.210241	0.230	1st
A3	Policy reforms	0.3922870	0.202358	0.512	4th
A4	Corruption free system	0.553201	0.30215	0.298	2nd
A5	Infrastructure development	0.250147	0.12021	0.600	5th

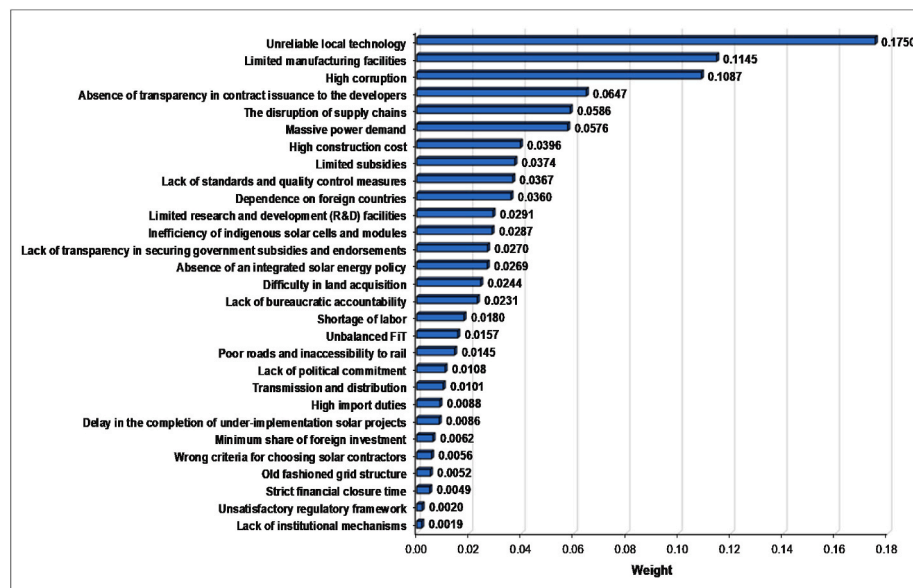


Fig. 10. The overall ranking of sub-barriers.

$$P(S_4 \leq S^{\max}) = 0.298 \quad (16)$$

$$(S_5 \leq S^{\max}) = 0.600 \quad (17)$$

The ordering of alternatives is determined by the available greyness values and is given in equation (18) as:

$$A_2 > A_4 > A_1 > A_3 > A_5 \quad (18)$$

The results of G-TOPSIS (reported in Table 5) show that R&D improvement is the top alternate solution to overcome solar energy development barriers by obtaining (0.230) weight. Corruption free system (0.298) ranks second in this category. Human resources availability, policy reformations, and infrastructure development are the least significant alternatives by obtaining weights of (0.401), (0.512), and (0.600), respectively.

4. Conclusions and policy propositions

This study evaluated natural resource volatility by assessing the influence of solar energy development barriers. Several barriers were initially scrutinized from the previous literature and then refined with the assistance of experts. By employing the AHP method, twenty-nine barriers within six major categories were finalized, including technological, transparency & accountability, COVID-19, financial, policy & regulatory, and infrastructure. The weights of all these barriers were obtained by performing a pairwise comparison matrix. The contributions of this paper begin with the results of the AHP. The findings suggest that the technological barrier ranks highest, followed by the transparency & accountability barrier, COVID-19, financial, policy & regulatory, and infrastructure barriers, respectively. The ranking of sub-barriers reveals that “unreliable local technology” is the sub-barrier with the highest weight. Further contributions are expanded by applying the G-TOPSIS approach, which has assisted authors in extending previous outcomes and prioritizing the top five alternatives for enhancing and validating the accuracy of research outcomes. Based on the findings of the G-TOPSIS approach, R&D improvement is deemed the top alternative, followed by a corruption-free system, human resources availability, policy reformations, and infrastructure development to address the challenges of overcoming natural resources volatility.

Among several natural resources, solar energy may be China's best energy option, as every year, the country receives massive sunlight. However, the country has significant challenges in attaining its solar power sector's maximum potential. Consequently, there is a substantial gap between the current and planned production of the solar industry. Research results revealed that R&D is an essential component of solar sector development. Therefore, it is proposed that R&D activities should be improved to minimize operating costs and improve domestic technology. Ample funds should be allocated to accelerate this process.

Dramatic changes caused by the novel COVID-19 pandemic have unprecedented implications for all industries, and the solar industry is no exception. The unavailability of human resources due to the countrywide lockdown is one of the solar industry's major challenges. The role of the government is critical in this regard. There is a need to establish crisis management strategies and formulate comprehensive policies for remote and hybrid working environments to respond to the current pandemic crises. The system should be free from corruption for the sustainable growth of the solar industry. It should be regulated to ensure that all solar development programs have appropriate controls and balances, reduce bureaucracy powers, and devise an effective tax mechanism. Developers should be given tax holidays for establishing new solar parks and take benefit of accelerated depreciation.

Without a sound policy framework, the idea of rapid implementation of solar power is unattainable. The government should develop a detailed national solar energy program in this respect. The existing regulatory framework is typically intended for thermal energy sources, and the solar industry has received minimal subsidies. In addition, solar park development and maintenance demand significant financial resources. The old policy framework needs an amendment by giving enough subsidies to the solar power sector. Similarly, the benchmark used for selecting developers for the solar mission program must be switched from minimum-maximum capability to financial status and track record basis. The solar energy infrastructure also requires major improvements. As the process of land acquisition for solar farms needs the approval of several government agencies and is always time-taking; therefore, the government should make efforts to make this process transparent and simplified. Reducing the involvement of too many government agencies and following the transparent ways of land acquisition would be handy in this regard. The safe transport of PVs and relevant material to the construction sites should be supported by adequate roads and rail. Finally, the inefficient and obsolete T&D mechanism should be upgraded to mitigate T&D losses.

Additionally, the study had limitations that might be addressed in future research. Other variables, such as the lack of infrastructure, danger, freedom, and insecurity, were omitted from our empirical investigation. These elements should be the focus of future research if they are to deliver intriguing results. During the COVID-19 shut down in China, data was obtained from specialists. The present COVID-19 conditions may restrict the generalizability of the findings. After the end of the pandemic, it is possible that the findings may change. In this context, it is recommended that future research revisit these barriers. This study employed a convenient sampling method, which has the potential to create sampling bias. To ensure inclusive study outcomes, academics may investigate additional sampling methods. In a similar manner, just fifteen experts' opinions were considered during the decision-making process. The inclusion of extra specialists may result in conclusions that are more exhaustive. Therefore, scholars should explore this significant subject in future research.

Author statement

Jinhao Liang: Visualization, Writing-Review & Editing. **Muhamad Irfan:** Project administration, Variable construction, Software, Writing-Original Draft, Writing-Review & Editing. **Muhammad Ikram:** Conceptualization, Data curation, Methodology, Formal analysis. **Dominik Zimon:** Writing-Review & Editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix-A

Table A1

Profile of experts.

No.	Participants	Experience	Qualification	Organization
1	Professor	26 years	PhD	Peking University
2	Associate professor	14 years	PhD	Tsinghua University
3	Lecturer	11 years	MPhil	Renmin University
4	Chairman & Managing Director	25 years	Masters	Energy Foundation China
5	Director	23 years	Masters	Energy Foundation China
6	Independent Director	27 years	Masters	Energy Foundation China
7	Executive Director	19 years	Bachelor	China Power Clean Energy Development
8	Deputy Director	20 years	Masters	China Power Clean Energy Development
9	Assistant Director	10 years	Masters	China Power Clean Energy Development
10	Chief Controller of Accounts	24 years	Bachelor	JinkoSolar
11	Joint Secretary	20 years	Bachelor	JinkoSolar
12	Section officer	15 years	Bachelor	JinkoSolar
13	Section officer	13 years	Bachelor	China Energy Investment
14	Marketing Executive	11 years	Bachelor	China Energy Investment
15	Director Operations	15 years	Bachelor	China Energy Investment

Appendix-B

Table B1

Pairwise comparison matrix of technological sub-barrier.

TECH	TECH -1	TECH -2	TECH -3	TECH -4	TECH -5	Wi	Rank
TECH-1	1	0.812	0.882	0.205	0.251	0.076	4th
TECH-2	1.232	1	2.101	0.167	0.193	0.094	3rd
TECH-3	1.134	0.476	1	0.164	0.328	0.075	5th
TECH-4	4.875	6.002	6.098	1	1.761	0.457	1st
TECH-5	3.982	5.181	3.049	0.568	1	0.299	2nd

CR = 0.032.

Table B2

Pairwise comparison matrix of transparency & accountability sub-barrier.

TRA	TRA -1	TRA -2	TRA -3	TRA -4	TRA -5	Wi	Rank
TRA-1	1	0.512	0.340	0.273	0.850	0.089	5th
TRA-2	1.953	1	0.568	0.238	1.547	0.141	3rd
TRA-3	2.943	1.762	1	0.468	3.014	0.249	2nd
TRA-4	3.658	4.202	2.138	1	2.936	0.418	1st
TRA-5	1.176	0.646	0.332	0.341	1	0.104	4th

CR = 0.021.

Table B3

Pairwise comparison matrix of COVID-19 sub-barrier.

COV	COV-1	COV-2	COV-3	COV-4	Wi	Rank
COV-1	1	1.236	3.683	4.892	0.403	2nd
COV-2	0.809	1	4.254	7.086	0.410	1st
COV-3	0.272	0.235	1	3.023	0.126	3rd
COV-4	0.204	0.141	0.331	1	0.060	4th

CR = 0.035.

Table B4

Pairwise comparison matrix of financial sub-barrier.

FIN	FIN-1	FIN-2	FIN-3	FIN-4	FIN-5	Wi	Rank
FIN-1	1	1.015	4.327	0.122	0.147	0.091	3rd
FIN-2	0.985	1	0.981	0.193	0.157	0.064	4th
FIN-3	0.231	1.019	1	0.132	0.169	0.051	5th
FIN-4	8.197	5.176	7.576	1	1.037	0.408	1st
FIN-5	6.803	6.369	5.917	0.964	1	0.386	2nd

CR = 0.077.

Table B5

Pairwise comparison matrix of policy & regulatory sub-barrier.

POLR	POLR-1	POLR-2	POLR-3	POLR-4	POLR-5	Wi	Rank
POLR-1	1	0.118	0.188	0.166	0.143	0.032	5th
POLR-2	8.457	1	4.762	3.217	2.659	0.441	1st
POLR-3	5.327	0.210	1	0.272	0.228	0.091	4th
POLR-4	6.034	0.311	3.678	1	0.470	0.177	3rd
POLR-5	7.017	0.376	4.381	2.127	1	0.257	2nd

CR = 0.076.

Table B6

Pairwise comparison matrix of infrastructure sub-barrier.

INF	INF -1	INF -2	INF -3	INF -4	INF -5	Wi	Rank
INF-1	1	0.120	0.192	0.166	0.143	0.033	5th
INF-2	8.346	1	4.651	3.106	2.548	0.435	1st
INF-3	5.216	0.215	1	0.280	0.234	0.092	4th
INF-4	6.023	0.322	3.567	1	0.496	0.180	3rd
INF-5	7.006	0.392	4.270	2.016	1	0.258	2nd

CR = 0.069.

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